

Latent Insider Information Shocks

Methodology, Variable Definitions, Data, and Empirical Results

Nima Taheri Hosseinkhani

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Abstract

This note documents the construction and empirical content of Latent Insider Information Shocks (LIIS), a transaction-level measure designed to separate economically informative insider filings from routine, planned, derivative-linked, compensation-related, or mechanical Form 4 activity. The empirical pipeline starts from WRDS Thomson Reuters Insiders, augments non-derivative transactions with derivative-table, amendment, role, and 10b5-1 support fields, and maps scored filings to CRSP PERMNO-month returns. The final production run scores 17,328,988 insider transactions, maps 11,624,607 transaction rows into the CRSP backtest, and produces a 390,627-row PERMNO-month LIIS panel over 203 monthly return periods. The main long-short portfolio formed on scaled LIIS earns 0.38 percent per month with a t-statistic of 1.94. The non-10b5 subset is stronger, earning 0.42 percent per month with a t-statistic of 2.20. The evidence supports a measurement interpretation: modern insider-filing information is not well summarized by raw net buying or selling alone; it is concentrated in transactions whose filing structure looks discretionary, material, and not mechanical. The current evidence is promising but not yet a top-journal causal claim; the Fama-MacBeth slope is weak in the current specification, and the signal weakens sharply in the liquid-large robustness screen.

1 Research Question and Contribution

The economic question is whether insider filings contain priced information after removing trades that are likely to be routine, pre-planned, derivative-linked, tax-related, amendment-related, or compensation-driven. The question is not whether insider trading can ever predict returns. That broad fact has a long literature, including Lakonishok and Lee (2001) and Cohen, Malloy, and Pomorski (2012). The sharper question is whether modern filing-level structure can improve measurement of which individual Form 4 transactions should count as information-bearing signals.

The project therefore treats insider filings as a measurement problem before it treats them as an asset-pricing signal. A raw Form 4 transaction is first converted into a transaction-level probability of being informative. The probability is learned from filing characteristics that distinguish open-market and material transactions from mechanical or planned transactions. These transaction scores are then aggregated into firm-month state variables and tested against future stock returns. This design is closest in spirit to the routine versus opportunistic distinction in Cohen, Malloy, and Pomorski (2012), but it differs in the unit of classification and in the information set: the object here is a transaction-level score that incorporates derivative companions, amendments, 10b5-1 indicators, filing lags, role information, and missingness structure.

The contribution is best stated as a measurement and implementation claim: modern insider filings contain useful return-relevant information, but the information is concentrated in transactions that survive a richer filing-structure filter. The strongest current evidence is the non-10b5 and

amendment-strict portfolio evidence. The weakest current evidence is the monthly cross-sectional Fama-MacBeth slope, which does not yet support a broad characteristic-pricing claim.

2 Data

The production run uses the following data sources.

Table 1: Data Inputs and Use in the LIIS Pipeline

Source	Tables or Files	Use
WRDS Thomson Reuters Insiders	Non-derivative Table 1; derivative Table 2; filing headers; 10b5-1 support; amendment support; company support	Defines insider transactions, filing dates, transaction dates, transaction codes, roles, ownership, 10b5-1 indicators, amendment links, and derivative companions.
CRSP monthly stock file	Monthly returns, prices, shares outstanding, market capitalization, and security identifiers	Maps LIIS to future stock returns and scales firm-month signals by lagged or current market capitalization.
CRSP daily stock file	Daily returns around filing dates	Constructs event-time return paths around filing availability.
CRSP names history	PERMNO, CUSIP, NCUSIP, share code, and valid name dates	Builds a date-valid CUSIP6-month to PERMNO bridge for insider transactions.
CRSP/Compustat Merged and Compustat	CCM link history; annual and quarterly fundamentals	Provides accounting and identifier support for the broader research design. The current headline tests use CRSP return joins and market-cap controls.
Kenneth French factor files	Monthly Fama-French three-factor series	Provides factor-loadings diagnostics for the LIIS long-short return.
TAQ entitlement probe	TAQ candidate schemas and small-access checks	TAQ schemas were visible, but sample queries returned permission errors. TAQ is therefore not used in the reported implementation-cost calibration.

The intended insider sample window is January 2004 through December 2025, with 2026 reserved as a future holdout. The realized return panel starts later after identifier mapping and return availability filters. The final backtest contains 203 monthly return periods and 390,627 PERMNO-month observations. All sample counts and coefficients in this note are taken from the current run manifests and summary tables, while all variable definitions are taken from the current pipeline code.

3 Transaction-Level Construction

Let j index an insider transaction, $i(j)$ the issuing firm, and f_j the filing date on which the transaction becomes available in the insider-filing data. Let τ_j denote the transaction date, q_j the reported share amount, p_j^{price} the reported transaction price, and a_j the acquired- or-disposed indicator. The trade sign is

$$s_j = \begin{cases} +1, & a_j = A \text{ or transaction code in } \{P, A\}, \\ -1, & a_j = D \text{ or transaction code in } \{S, D\}, \\ \text{missing}, & \text{otherwise.} \end{cases} \quad (1)$$

The signed and absolute dollar values are

$$V_j = s_j |q_j| |p_j^{\text{price}}|, \quad |V_j| = |V_j|. \quad (2)$$

The filing lag is

$$\ell_j = f_j - \tau_j, \quad (3)$$

measured in calendar days. In the predictive feature vector, the lag is clipped to the interval $[-30, 365]$, with missing lags assigned a separate large value.

Transaction code families are defined from the Form 4 code. Open-market trades are codes P and S . Mechanical-code transactions include grants or awards, option or derivative exercises, tax or disposition codes, gifts, and other non-standard code families. The indicator for a mechanical transaction is

$$M_j = \mathbf{1}\{\text{grant or award}\} \vee \mathbf{1}\{\text{option or derivative code}\} \\ \vee \mathbf{1}\{\text{tax or disposition code}\} \vee \mathbf{1}\{\text{gift or other code}\}. \quad (4)$$

Role indicators identify CEO/CFO, director, and officer language in the reporting person title or role-code fields. The 10b5-1 flag is constructed from both the main transaction record and the support table. Derivative companions are identified by grouping derivative Table 2 records by filing accession, person, and security identifiers. Let D_j be the derivative companion count. The indicator

$$H_j^{\text{deriv}} = \mathbf{1}\{D_j > 0\} \quad (5)$$

marks transactions that have at least one derivative companion record. Amendment status is identified from amendment support tables or, when the support table is absent, from the amendment flag in the main insider record.

The routine-or-mechanical flag is

$$R_j = M_j \vee B_j^{10b5} \vee H_j^{\text{deriv}} \vee A_j^{\text{amend}}, \quad (6)$$

where B_j^{10b5} is the 10b5-1 indicator and A_j^{amend} marks an amended record. The transaction-level weak label used for model training is

$$Y_j = \mathbf{1}\{\text{open-market}_j\} \cdot \mathbf{1}\{B_j^{10b5} = 0\} \cdot \mathbf{1}\{R_j = 0\} \cdot \mathbf{1}\{|V_j| > Q_{0.60}(|V|)\}. \quad (7)$$

This label is not a legal definition of informed trading. It is a noisy supervision device: large, open-market, non-10b5, non-mechanical transactions are treated as candidate informative events, while routine or mechanical transactions are downweighted.

4 Predictive Model and Score

The transaction-level feature vector X_j includes

$$X_j = (\log(1 + |V_j|), \ell_j^{\text{clip}}, \log(1 + \min\{1000, |q_j|/|h_j|\}), \log(1 + D_j), \quad (8)$$

$$\mathbf{1}\{f_j \geq \text{April 1, 2023}\}, \mathbf{1}\{\text{open-market}\}, M_j, B_j^{10b5}, \mathbf{1}\{\text{CEO/CFO}\}, \quad (9)$$

$$H_j^{\text{deriv}}, A_j^{\text{amend}}, \mathbf{1}\{\text{missing price}\}, \mathbf{1}\{\text{missing shares}\}, \quad (10)$$

$$\mathbf{1}\{\text{tax or disposition code}\}, \mathbf{1}\{\text{option or derivative code}\}. \quad (11)$$

Here h_j is shares held after the transaction when available. The 2023 indicator reflects the modern 10b5-1 disclosure era after the SEC's 2022 amendments to insider-trading plan disclosures.

Three model families are estimated. First, rules-only scoring provides a transparent baseline. Second, Elastic Net logistic regression provides a sparse linear benchmark. Third, a LightGBM classifier provides the main tabular model. A PyTorch neural-network branch is trained on the same transaction-level features as a GPU extension. The final score is a blend:

$$\hat{p}_j = 0.70 \hat{p}_j^{\text{LGBM}} + 0.30 \hat{p}_j^{\text{MLP}}. \quad (12)$$

In the production environment, LightGBM's GPU tree learner was unavailable in the installed build, so the LightGBM branch ran on CPU. The PyTorch branch did use the A100 GPU and trained on 3,000,000 transaction rows for three epochs.

The transaction-level signed information value is

$$Z_j = s_j \hat{p}_j |V_j|. \quad (13)$$

This object is positive for predicted-informative purchases, negative for predicted-informative sales, and close to zero for small or low-probability transactions.

5 Firm-Month LIIS Variables

Let $\mathcal{J}_{i,t}$ be the set of insider transactions for firm i filed in month t and mapped to a valid CRSP PERMNO. The main firm-month signal is

$$\text{LIIS}_{i,t}^{\text{net}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j. \quad (14)$$

The buy and sell components are

$$\text{LIIS}_{i,t}^{\text{buy}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j \mathbf{1}\{Z_j > 0\}, \quad \text{LIIS}_{i,t}^{\text{sell}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j \mathbf{1}\{Z_j < 0\}. \quad (15)$$

Role- and mechanism-specific variants are constructed by applying indicator filters before aggregation:

$$\text{LIIS}_{i,t}^{\text{CEO/CFO}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j \mathbf{1}\{\text{CEO/CFO}_j\}, \quad (16)$$

$$\text{LIIS}_{i,t}^{\text{nonplan}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j \mathbf{1}\{B_j^{10b5} = 0\}, \quad (17)$$

$$\text{LIIS}_{i,t}^{\text{open}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j \mathbf{1}\{\text{open-market}_j\}, \quad (18)$$

$$\text{LIIS}_{i,t}^{\text{amend-strict}} = \sum_{j \in \mathcal{J}_{i,t}} Z_j \mathbf{1}\{A_j^{\text{amend}} = 0\}. \quad (19)$$

Each signal is scaled by market capitalization:

$$\widetilde{\text{LIIS}}_{i,t} = \frac{\text{LIIS}_{i,t}^{\text{net}}}{\text{MCap}_{i,t-1}}, \quad (20)$$

using lagged market capitalization when available and current market capitalization otherwise. The scaled variables enter portfolio sorts and cross-sectional regressions.

The portfolio construction sorts stocks into deciles within each month using $\widetilde{\text{LIIS}}_{i,t}$. The long-short return is

$$R_{t+1}^{10-1} = \frac{1}{N_{10,t}} \sum_{i \in D_{10,t}} R_{i,t+1} - \frac{1}{N_{1,t}} \sum_{i \in D_{1,t}} R_{i,t+1}. \quad (21)$$

Cumulative performance is computed as $\prod_{\tau \leq t} (1 + R_{\tau}^{10-1}) - 1$.

The Fama-MacBeth-style cross-sectional regression is

$$R_{i,t+1} = \alpha_t + \beta_t \widetilde{\text{LIIS}}_{i,t} + \gamma_t \log(\text{MCap}_{i,t}) + \varepsilon_{i,t+1}, \quad (22)$$

estimated separately by month. The reported slope is the time-series mean of β_t divided by its time-series standard error. Factor diagnostics estimate

$$R_t^{10-1} = \alpha + \beta_M \text{MKT}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \eta_t \quad (23)$$

against monthly Fama-French three-factor returns.

6 Variable Glossary

Table 2: Main Variables Created by the Pipeline

Variable	Definition
s_j	Transaction sign. Purchases/acquisitions are +1; sales/dispositions are -1.
V_j	Signed dollar value, $s_j q_j p_j^{\text{price}} $.
$ V_j $	Absolute transaction dollar value.
ℓ_j	Filing lag in days, filing date minus transaction date.
<code>is_open_market</code>	Indicator for open-market purchase or sale codes P and S .
<code>is_mechanical_code</code>	Indicator for grant, award, derivative exercise, tax-withholding, gift, or other mechanical transaction-code families.
<code>is_10b5</code>	Indicator that the transaction is marked as related to a 10b5-1 plan in the main or support table.
<code>has_derivative_companion</code>	Indicator that derivative Table 2 contains at least one companion record linked to the same filing/person/security combination.
<code>is_amended_record</code>	Indicator that the transaction belongs to or is linked to an amendment record.

Variable	Definition
R_j	Routine-or-mechanical flag combining mechanical codes, 10b5-1 status, derivative companions, and amendment status.
Y_j	Weak informative label for model training: large open-market, non-10b5, non-routine transactions.
$\hat{p}_j$	Predicted probability that transaction j is economically informative.
Z_j	Signed information value, $s_j \hat{p}_j V_j $.
$\text{LIIS}_{i,t}^{\text{net}}$	Sum of Z_j across transactions filed by firm i in month t .
$\widetilde{\text{LIIS}}_{i,t}$	Market-cap-scaled net LIIS.
$\text{LIIS}_{i,t}^{\text{nonplan}}$	LIIS using only transactions without a 10b5-1 indicator.
$\text{LIIS}_{i,t}^{\text{amend-strict}}$	LIIS excluding amended records.
<code>liis_cluster</code>	Maximum of distinct buy-side and sell-side insider owner counts in a firm-month.
<code>liis_disagreement</code>	Entropy-style disagreement measure based on the share of distinct buy-side versus sell-side owners.
R_{t+1}^{10-1}	Equal-weight return of LIIS decile 10 minus LIIS decile 1 in the next month.

The owner-count variables are included because the pipeline creates them for cluster and disagreement diagnostics. In the current mapped panel, the owner count fields do not contribute meaningful variation, so they should not be used as evidence for the headline result until the owner identifier feed is audited.

7 Empirical Results

7.1 Transaction scoring and model interpretation

The model-training run scores 17,328,988 transactions and uses 13,639,115 transactions in the training window. Elastic Net logistic regression completes successfully, the LightGBM tabular model completes on CPU, and the PyTorch GPU MLP branch completes on the A100. Table 3 summarizes the model status.

Table 3: Model Variants

Model	Role	Status
Rules only	Transparent baseline	Available
Elastic Net logistic	Sparse benchmark	Ok
LightGBM	Main tabular workhorse	Ok, CPU tree learner
Torch GPU MLP	GPU sequence/tabular extension	Ok
Blend	Production score	LightGBM CPU plus torch GPU MLP

The SHAP summary in Figure 1 indicates that the most important stage-one inputs are ab-

solute dollar value, open-market status, and derivative companion structure. This interpretation is economically coherent: the score puts weight on material discretionary trades and discounts transactions that look mechanical or derivative-linked.

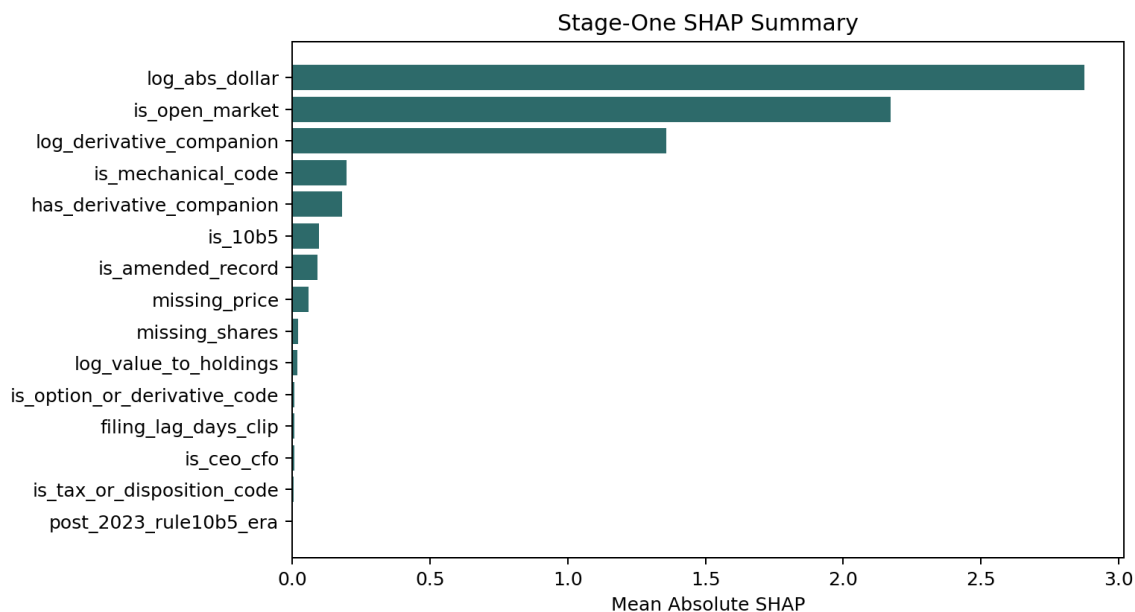


Figure 1: Stage-One SHAP Summary. The figure reports mean absolute SHAP values for the LightGBM stage-one model.

7.2 Portfolio returns

The main LIIS decile portfolio produces 203 monthly long-short returns. The average monthly return is 0.35 percent in the realized decile table and 0.38 percent in the robustness recomputation, with a robustness t-statistic of 1.94. The approximate annual return is 4.26 percent, the annual volatility is 9.87 percent, and the approximate Sharpe ratio is 0.43. The maximum drawdown is -25.63 percent.

Table 4: Performance Summary

Series	Months	Mean monthly	Annual return	Annual vol.
Long-short decile 10 minus decile 1	203	0.00355	0.04258	0.09873

The maximum drawdown for this series is -25.63 percent.

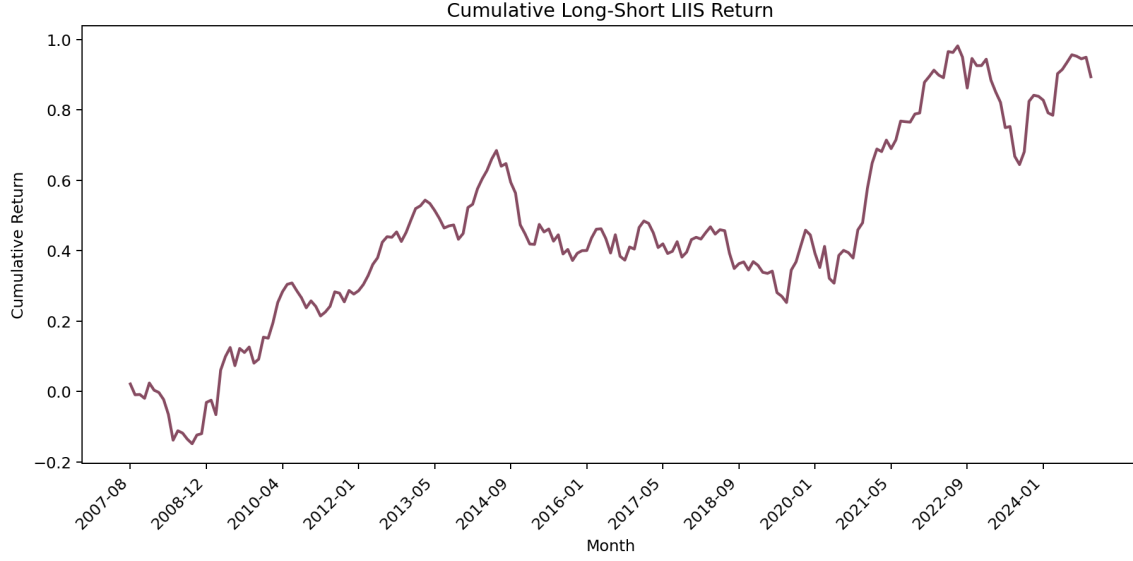


Figure 2: Cumulative Long-Short LIIS Return. The line plots cumulative wealth from the equal-weight decile 10 minus decile 1 return.

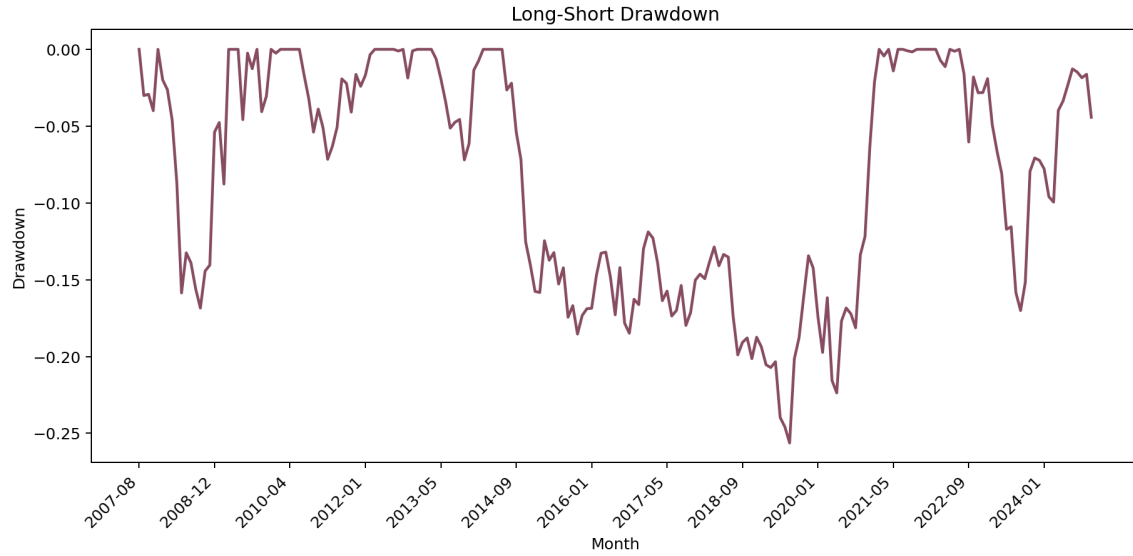


Figure 3: Long-Short Drawdown. The figure reports drawdowns for the same long-short return series.

7.3 Robustness

Table 5 reports the robustness battery. The strongest variant is the non-10b5 signal, with a mean monthly long-short return of 0.42 percent and a t-statistic of 2.20. The amendment-strict version is close to the main signal, with a t-statistic of 1.92. The open-market-only version remains positive but weaker. The liquid-large screen is weak, which is an important scope condition for publication claims and implementation interpretation.

Table 5: Robustness Summary

Test	Months	Mean long-short	t-statistic
Main LIIS net	203	0.00380	1.94
Buys only	203	0.00217	1.23
Sells only	203	-0.00060	-0.31
CEO/CFO only	203	0.00221	1.12
Non-10b5 only	203	0.00424	2.20
Open-market only	203	0.00282	1.50
Amendment strict	203	0.00381	1.92
Liquid-large 70 percent	203	0.00019	0.09
Placebo shuffle within month	203	0.00134	1.28
Transaction-date alignment	251	0.00484	2.88

7.4 Event-time validation

The event-time validation groups filing-date events by the event-level LIIS score and computes cumulative average daily returns over business days -5 to $+5$ around filing availability. The current event-time exhibit uses CRSP daily returns and should be interpreted as an event-time return validation rather than a fully benchmark-adjusted abnormal-return design. The run includes 939,844 event-window observations across 10 event deciles.

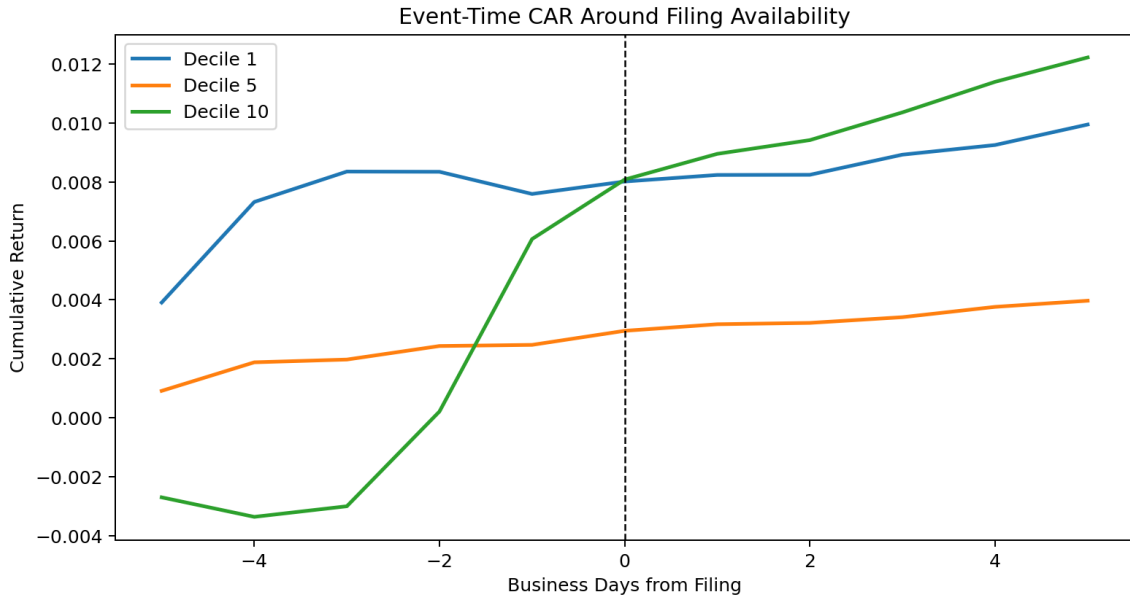


Figure 4: Event-Time Cumulative Returns Around Filing Availability. The figure plots selected LIIS event deciles from business days -5 to $+5$.

7.5 Factor and cross-sectional pricing diagnostics

The factor-loading diagnostic regresses the LIIS long-short return on the Fama-French market, SMB, and HML factors. The estimated monthly intercept is 0.36 percent over 203 months. The market loading is small, while the HML loading is positive.

Table 6: Fama-French Factor Loading Diagnostic

Term	Loading	Months
Alpha	0.00361	203
Mkt-RF	0.01103	203
SMB	0.03633	203
HML	0.10474	203

The current Fama-MacBeth-style cross-sectional diagnostic is weaker. The time-series mean coefficient on scaled LIIS is -0.0070 with a t-statistic of -0.28 over 203 months. This result argues against making a broad characteristic-pricing claim at this stage. The portfolio evidence is promising, but a top-journal version of the paper should either strengthen the cross-sectional specification or frame the evidence as a portfolio/measurement result rather than as a fully established priced characteristic.

Table 7: Fama-MacBeth-Style Monthly Cross-Section

Term	Mean	t-statistic	Months
Alpha	0.02332	1.76	203
Scaled LIIS	-0.00702	-0.28	203
Log market capitalization	-0.00071	-1.33	203

8 Interpretation and Publication Boundary

The evidence supports a restrained conclusion. LIIS improves the economic description of the insider-filing stream by distinguishing material, discretionary-looking transactions from planned or mechanical transactions. The positive non-10b5 and amendment-strict results are consistent with the view that modern filing structure matters. The SHAP evidence supports the same interpretation: the model’s strongest inputs are dollar size, open-market status, and derivative companion structure.

The evidence does not yet support an unqualified top-journal claim that LIIS is a new priced characteristic. The liquid-large robustness result is weak, and the simple Fama-MacBeth slope is not supportive in the current specification. The right next step is a benchmarked second pass: compare LIIS directly against the routine/opportunistic classification in Cohen, Malloy, and Pomorski (2012), show incremental value beyond simple open-market buys and sells, and split the sample around the modern 10b5-1 disclosure regime introduced by the SEC’s 2022 amendments. If those gates pass, the paper’s strongest claim would be that modern transaction-level Form 4 structure changes which insider filings remain informative after routine and planned trading are removed.

9 Replication Notes

The full production pipeline requires WRDS access. The public repository contains the source code, configuration, summary tables, figures, and tests, but does not redistribute raw WRDS extracts, processed proprietary panels, logs, manifests, credentials, or model binaries. A full rerun proceeds through five phases: extract, features, models, backtest, and figures. The production run used restartable year-sharded extracts and cached successful WRDS pulls before downstream modeling.

References

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